GGG-S-60C
February 9, 1978
SUPERSEDING
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April 14, 1971

FEDERAL SPECIFICATION

SAWS, BUTCHER'S; AND BLADES, BUTCHER'S HANDSAW

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two sizes of butcher's handsaws and related blades.

1.2 Classification. The butcher's handsaws and related blades covered herein shall be furnished as complete handsaws with blades, or blades only (see 6.2). The handsaws and the blades shall be of the following sizes, as specified (see 6.2).

Item I - Handsaw complete with blade

Size 1 - 21 inches (533 mm), overall length
Size 4 - 31 inches (787 mm), overall length

Item II - Blade only.

Size 1 - 14 inches (356 mm), nominal length
Size 2 - 18 inches (457 mm), nominal length
Size 4 - 24 inches (610 mm), nominal length

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing

FSC 7330

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Federal Standard:

FED-STD-123 - Marking for Shipment (Civil agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation-Packaging, Methods of
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with the specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for approval shall apply:

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National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The butcher's saw and blades delivered under this specification shall be the manufacturer's commercial standard product except for any changes necessary to comply with specification requirements.

3.2 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial butcher's saw provided the completed item complies with all provisions of this specification (see 6.4).

3.3 Design and construction. The complete handsaw shall consist of an open steel frame with a handle and a removable blade with either hole or pin ends. A device for applying tension to the blade shall also be provided and shall be a trigger type or wingnut as specified (see 6.2). The handsaw shown in Figure 1 is illustrative only and is not intended to preclude any commercial design otherwise conforming to specification requirements. The handsaw and blade shall conform to the dimensions shown in table I and II, respectively.

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TABLE I. Handsaw dimensions

Size	Overall length ± 1 (25) inches (mm)	Minimum depth from teeth of blade to inside edge of frame inches (mm)	Nominal cross section width of frame inches (mm)	Nominal frame thickness inches (mm)
1	21 (533)	4 (102)	1/2 (13)	3/16 (5)
4	31 (787)	4-3/8 (111)	5/8 (16)	3/16 (5)

TABLE II. Blade dimensions

Size	Length of cutting edge ± 1/2 (13) inches (mm)	Nominal blade width inches (mm)	Teeth points per inch	Minimum blade thickness inches (mm)	1/ Minimum effective cutting length inches (mm)
1	14 (356)	1/2 (13)	11	0.022 (.56)	12-1/2 (318)
2	18-1/2 (470)	1/2 (13)	11	0.022 (.56)	17 (432)
4	24 (610)	1/2 (13)	11	0.022 (.56)	22-1/2 (572)

1/ See 6.3.

3.3.1 Frame. The handsaw frame shall be nonadjustable and shall be fabricated from steel bar stock. The frame, with handle, shall be capable of withstanding the tests specified in 4.3.1 without failure of any parts. When tested as specified in 4.3.1 (a), the portion of the frame between the blade holding devices shall not contract more than 1/4 inch (6.4 mm). The frame shall have no permanent set when tested as specified in 4.3.1 (b).

3.3.2 Handle. The handle shall be made from high impact plastic or any clear, straight-grained, thoroughly seasoned hardwood as specified (see 6.2). Handle edges shall be rounded. The handle shall be secured to the frame with a minimum of three oval head screws. The undersurface of the screw heads shall be flush with the surface of the handle.

3.3.3 Blade. The blades shall be steel, heat treated and tempered over the entire length and width so that the teeth area has a Rockwell hardness between C65 and C70 (see 4.3.2). The teeth of the blades shall be evenly spaced and of uniform height.

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3.3.3.1 Blade holding devices. The blade holding devices shall prevent rotation or twisting of the blade and shall be capable of applying a minimum tension of 20 pounds (9 kg) to the blade when tested as specified in 4.3.3. The blade holding device shall be attached to the frame at the handle end of the handsaw. The blade bearing surfaces shall be smoothly finished to provide a flat bearing surface for the blade. Provision shall be made to prevent the bladeholding devices from falling out of the frame in the event of blade breakage.

3.4 Finish.

3.4.1 Frame and blade. Each frame and blade shall be finished in accordance with the manufacturer's standard commercial practice, except cadmium plating shall not be used.

3.4.2 Handle. Each wood handle shall have a natural lacquer finish.

3.5 Marking. Each handsaw frame shall be permanently and legibly marked with trade name or trademark of such known character that the source of manufacture may be readily determined.

3.6 Workmanship. There shall be no burrs, rough or sharp edges, or corners that may cause injury to personnel.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Sampling for inspection shall be made in accordance with MIL-STD-105, unless otherwise indicated hereinafter.

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4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with the requirements of referenced specifications, standards, and drawings unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item inspection. The sample unit shall be one complete handsaw or blade, as applicable.

4.2.2.1 Visual examination. Examination shall be made of the handsaw and blade, as applicable, for the defects listed in table III. The inspection level shall be II, with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE III. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not natural lacquer finish on wood handle.		X
Construction and workmanship	Frame not one piece.	X	
	Burrs, rough or sharp edges, or corners on frame.	X	
	Handle not straight-grained if wood.		X
	Handle if plastic not high impact plastic.	X	
	Handle edges not rounded.		X
	Handle not secured to frame with at least three screws.	X	
	Screws not oval head.		X
	Undersurface of screw not flush with surface of handle.	X	
	Blade has no holes or pins.	X	
	Rear blade holding device not either wingnut type or trigger type.	X	
	Blade holding devices not integral part of frame or not permanently attached to frame or combination of both.	X	
	Blade bearing surfaces of blade holding devices not flat.		X
Marking	Missing, incomplete, illegible, not in proper location.		X

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4.2.2.2 Dimensional examination. An examination shall be made of the handsaw and blade, as applicable, for defects in dimensions. Any dimension not within specified requirements shall be classified a defect. The inspection level shall be S-2 with an AOL of 4.0, expressed in terms of defects per hundred units.

4.2.2.3 End product testing. The handsaw and blade, as applicable, shall be tested as specified in 4.3.1, 4.3.2, and 4.3.3. The inspection level for each test shall be S-4 with an AOL of 4.0, expressed in terms of defects per hundred units.

4.2.3 Inspection of packaging. An examination shall be made to determine that preservation-packaging, packing, and marking comply with the requirements of section 5. Defects shall be as indicated in table IV. The lot shall be the number of shipping containers offered for inspection at one time. The sample unit shall be one shipping container fully prepared for delivery except that it need not be closed. The inspection level shall be S-2 with an AOL of 4.0, expressed in terms of defects per hundred units.

TABLE IV. Inspection of packaging

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, or sequence; improperly applied.
Preservation	Preservative improperly applied or missing.
Materials	Component missing, damaged, or not as specified.
Workmanship	Bulged or distorted containers.
Weight	Net weight exceeds requirements.

4.2.3.1 Inspection of shipping containers. Inspection for closure, waterproofing, and reinforcing shall be in accordance with the quality assurance provisions contained in the appendix to PPP-B-636.

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4.3 Tests.

4.3.1 Load test. The distance between the blade holding devices shall be measured and recorded while the frame is subjected to a 20 pound (9.1 kg) compressive load applied between the blade holding devices. The load test shall then proceed as follows:

(a) The compressive load on the frame shall be increased from 20 to 100 pounds (9.1 to 45.4 kg) and held at 100 pounds (45.4 kg) while the distance between the bladeholding devices is measured and compared to the original measurement to determine compliance with 3.3.1. Any contraction over 1/4 inch (6.4 mm) or parts breakage shall constitute failure of this test.

(b) The compressive load in the frame shall then be decreased from 100 to 20 pounds (45.4 to 9.1 kg) and held at 20 pounds (9.1 kg) while the distance between the bladeholding devices is measured and compared to the original measurement to determine compliance with 3.3.1. Any permanent set or parts breakage shall constitute failure of this test.

4.3.2 Hardness. Blades shall be tested for hardness within 1/8 inch (3.2 mm) of base of tooth to determine compliance with 3.3.3. Nonconformance shall constitute failure of this test.

4.3.3 Tension. The tension applied by the rear bladeholding device shall be measured to determine compliance with the minimum tension requirement of 3.3.3.1. This measurement shall be obtained by attaching a scale or other suitable measuring device between the blade holding devices and tightening the rear blade holding device. Noncompliance with tension requirements shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be level A or commerical, as specified (see 6.2).

5.1.1 Level A preservation and packaging.

5.1.1.1 Disassembly. Each handsaw blade shall be removed and replaced finger tight, with its cutting edge toward the frame.

5.1.1.2 Preservation. All uncoated, unplated, ferrous metal surfaces of the saw blade (before reassembly) shall be cleaned process C-1, thoroughly dried, and coated with type P-14 preservative in accordance with MIL-P-116.

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5.1.1.3 Packaging. Each complete handsaw, or six handsaw blades, of one size only, prepared as specified in 5.1.1.1 and 5.1.1.2, shall be completely wrapped in greaseproof barrier material conforming to type II, grade A, class 1 or 2 of MIL-B-121. The wrap shall be secured with twine made of jute, sisal, manila, or cotton, or tape conforming to type III, grade B of PPP-T-45, or heat sealed when heat sealable barrier material is used.

5.1.2 Commercial. The handsaws or handsaw blades shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling and storage.

5.2 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A. Complete handsaws of one size only, or handsaw blades of one size only, preserved and packaged as specified in 5.1 shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each container shall be fitted with a liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636. The weight of contents of each shipping container shall not exceed 65 pounds (29.5 kg).

5.2.2 Level B. Complete handsaws of one size only, or handsaw blades of one size only, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF, (variety SW), or type SF, class domestic, grade 200 of PPP-B-636. Each shipping container shall be securely closed in accordance with method II as specified in the appendix of the container specification. The weight of contents of each shipping container shall not exceed 65 pounds (29.5 kg).

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

5.2.3 Commercial. The handsaws or handsaw blades shall be packed in fiberboard boxes to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

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5.3 Unitization. When shipments to Government depots are full car or truckload, the shipping containers shall be unitized for shipment and handling in accordance with normal commercial practice. The unitized load shall not exceed 2500 pounds (1134 kg) in weight, 63 inches (1600 mm) in height, 56 inches (1422 mm) in length and 45 inches (1143 mm) in width.

5.4 Marking. Marking shall be in accordance with 5.4.1 or 5.4.2 as specified (see 6.2).

5.4.1 Civil agencies. Packages, shipping containers and unitized loads (when applicable) shall be marked in accordance with FED-STD-123.

5.4.2 Military requirements. In addition to any special marking required the contract or purchase order, all interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The butcher's handsaws and blades are intended for use in meat-cutting plants and commissaries.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Item and size required, as applicable (see 1.2).
(c) Blade holding device required (see 3.3).
(d) Handle material required (see 3.3.2).
(e) For item II, blade style required (with hole or pin ends).
(f) Selection of applicable levels of preservation-packaging, and packing (see 5.1 and 5.2).
(g) When weather-resistant class fiberboard shipping containers are required for level B shipment (see 5.2.2.1).
(h) Marking required (see 5.4).

6.3 Minimum effective cutting length is defined as the distance from inside of frame at one end to inside of frame at opposite end along tooth edge of blade with blade in sow.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.2).

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6.5 Metric equivalents. Metric equivalents, indicated in parenthees throughout this document, are based on practice, conversion factors and symbols specified in ASTM E 380 Standard for Metric Practice and are for information only. In each instance, the value stated in US customary units shall be controlling.

Custodians:

Army - GL
Navy - SA
Air Force - 99

Review activities:

Army - MD
Navy - MS, CG
Air Force - 45

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

GSA - FSS

Project No. 7330-0437

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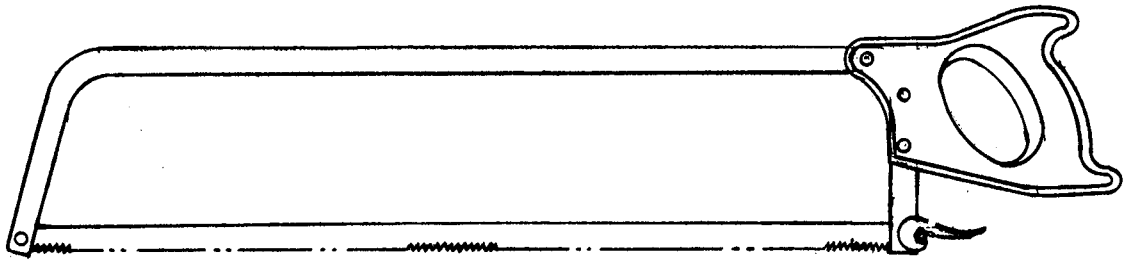


FIGURE 1. SAW, BUTCHER'S; AND BLADE, BUTCHER'S HANDSAW

2-9-191.B

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain copies of documents referenced herein. Price 70 cents each.

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